Robot Design Speech

Write it in your own words. Facts, numbers and reasons.

Updated 2026-11-12. The blue handwriting-style text is what the team wrote on 2026-11-05. Each section has a review box that says whether it is at Developing, Accomplished or Exceeds so far. **16 days to the tournament.**

Name: _____ **Date:** _____

At judging we give a **5-minute robot design explanation** with no game field, then judges ask questions for 5 more minutes. These pages guide you through each part of the speech. They give you sentence starters, not the answers. **You** write what we really did, so it sounds like us.

What to do

1. **Read** the gold box on each page. It tells you what judges listen for.
2. **Write** an answer under each ✎ starter. Use the reference pages (meeting notes, design log, test results) to find facts.
3. **Add numbers.** How many runs? How far off? How many out of 10?
4. **Don't know?** Write **[ASK: name]** and we will track it down.
5. **Practice out loud.** About 600 to 700 words is 5 minutes. Speak slowly.

What makes a speech score high?

- **Say why.** "We changed it because ..." beats "We changed it."
- **Say what is new.** Tell the judges what we changed from the original build.
- **Tell the test story:** we tested, found a problem, changed it, tested again, it got better.
- **Name everyone.** All nine of us speak and all nine of us did real work.
- **Be proud.** Smile. Celebrate what we did.

The robot design process we follow

Identify → Design → Create → Iterate (test and improve) → Communicate what we learned. The seven speech sections follow this order.

What judges listen for

Judges tick one box per row. Clear evidence is **Accomplished (3)**. To score **Exceeds (4)** the judges must write down how we exceeded, so give them something to write.

Area	Accomplished: clear evidence of ...	Exceeds: give them more
IDENTIFY	Mission strategy. Use of building and coding resources.	Why we chose missions in that order. Which resources, and what we took from each. Who we asked for help.
DESIGN	All members contributed ideas. Building and coding skills in all members.	Name **every** teammate and one thing each did. Who learned a new skill?
CREATE	Clear explanation of attachments and their purpose. Clear explanation of code and sensors.	What we **changed from the original** and why. Code explained **step by step**. What each sensor does for us.
ITERATE	Repeated testing. Improvements based on testing.	**Numbers** (how many runs, how far off). A story: test, problem, change, test again.
COMMUNICATE	Detailed process and lessons learned. Pride and enthusiasm.	Advice for another team. A mistake we learned from. Smiles and teamwork.

Who says what

Everyone speaks: Sylvie, Brynn, Jack, Fedor, Ira, Jordan, Ella, Alexis, Amelia. Times add up to about 4 minutes 40 seconds. You can split a section between two people.

Section	Time	Speaker(s)
1. Opening and mission strategy	0:30	
2. Choosing and building our robot	0:40	
3. Our attachments	1:00	
4. Our code and sensors	0:50	
5. Testing and improving	0:50	
6. What we learned	0:25	
7. Teamwork and closing	0:25	

Where our speech is now

Levels follow the rubric: Beginning (1), Developing (2), **Accomplished (3)**, Exceeds (4). This is a review of what is written so far, not a score from judges.

Area	Level now	What we have	To go up a level
IDENTIFY	DEVELOPING	Reasons for the missions and their order. Resources named.	Name the missions. Say what we took from each resource and who helped.
DESIGN	DEVELOPING	Four teammates named in the forklift story. Squaring shows coding skill.	Show building **and** coding skills for all 9, and everyone's ideas.
CREATE	DEVELOPING	Squaring, sensors, the lifting arm and the Mission 1 attachment: clear (Accomplished). Gyro Move, Kp and Ki, yaw, ramp and cm: simple explanation written, needs the why and numbers. Forklift: purpose and changes missing.	Every attachment: purpose, what changed, why. Explain the gyro code step by step.
ITERATE	ACCOMPLISHED	Story 1 (Kp and Ki tuning) has repeated tests, numbers and a change from the results. Story 2 (timer blocks) has no numbers yet.	Add the before and after number to story 1, and numbers to story 2. Show a retest result.
COMMUNICATE	NOT WRITTEN YET	Lessons and closing not written. The speech has not been practiced out loud.	Write sections 6 and 7, then rehearse. Smile and show pride.

Do these first today

1. **Name the missions** we attempt, in order (section 1).
2. **Finish the wheel story:** what we changed it to and why (section 2).
3. **Forklift:** what it does for Mission 9 and what we changed from the original (section 3).
4. **Test stories (section 5):** add a before and after number to story 1 and real numbers to story 2.
5. **Code:** how Gyro Move, Kp and Ki and yaw turns work (section 4, part 2).
6. **Sections 6 and 7** and the who-says-what table, so all 9 of us speak.

How to reach Exceeds

The judges must write down how we exceeded. Use **numbers**, tell what is **new** compared with the original, and show a **test that led to a change**.

1 Opening and mission strategy — IDENTIFY

WHAT JUDGES WANT
How we decided which missions to try, and which resources and people we learned from. Name at least three resources and say what each gave us.

Our run order: 1. Mission 9 Research Platform → 2. Mission 1 Drone Survey → 3. Mission 3* Exploding Seeds → 4. Mission 15 Biocentric Architecture (city dock: garden skylight and nesting canopy) → 5. Mission 6 Leafcutter Frenzy → 6. Mission 11 Window to the Past. *number to confirm

✎ **Hello, we are team _____ and today we will explain how we designed our robot. Our strategy is to attempt these missions:**

Hello, We're Look it's a Giraffe! and today we will explain how we designed our robot. Our strategy is to attempt missions that got us lots of points and were relatively easy. We looked at all the different missions and ranked them by two things: they needed to be able to get us lots of points, so we have a higher chance of doing better. The other way was how easy it is to achieve the mission, so we can go fast and get more missions done.

✎ **We chose those missions because ...** (points? location? one attachment for several?)

✎ **We decided to do them in this order because ...**

We did the missions in that order because we wanted to get from one side to the other just once to save time.

✎ **To learn building and coding we used these resources: (bot-fluencers, conference sessions, SPIKE app, build instructions)**

When building and coding the robot we used bot-fluencers, conference sessions, SPIKE app and build instructions.

✎ **From _____ we learned _____, and we used it for _____.**

✎ **When we got stuck we asked _____ for help with _____.**

DEVELOPING **Review against the rubric (2026-11-12)**

Good: Two clear reasons for picking missions (lots of points, easy to do), a reason for the order (cross the field once), and resources named.

Missing: The speech does not **name the missions** yet, and "We chose those missions because" is blank. The resources are a list with no detail: what did we take from each, and who did we ask for help?

Next: Accomplished: say the missions in order (see the run order above) and give 3 specific resource examples. **Exceeds:** add numbers (points per mission, our 185 point run) and how the plan changed after testing.

2 Choosing and building our robot

IDENTIFY + DESIGN

WHAT JUDGES WANT

A reason for our base robot, not just its name. Facts: we scored 4 designs on 6 criteria about match performance, left cost out on purpose, and HummerOne scored 23 out of 30. See log entries 1, 2, 3, 5 and 9.

✎ **We looked at __ designs: _____. We scored them on __ things: _____.**

We looked at 4 different designs: HummerOne, Comp Bot, GummyBears / Droid Bot M and StarterBot. We scored them on six criteria, all about how the robot performs in a match: archetype match to proven competitive designs, attachment swap speed at base, squaring (flat back and sides), stability under load, maneuverability in tight spaces and drive repeatability (for PID code).

✎ **We left out _____ on purpose because ...**

✎ **The highest score was _____ with __ out of __. We chose it because ...**

✎ **After building it we noticed this problem:** (front wheels? something sticking out?)

After we finished building the HummerOne we noticed the front wheels would slide and create friction, which made it harder to move straight.

✎ **So we changed _____ to _____ because ... (then again, if it changed twice)** (what was wrong with each version?)

We changed the front wheels to two mini ball casters, which helped cause less friction, which helped the robot move straight.

DEVELOPING **Review against the rubric (2026-11-12)**

Good: Names all 4 designs and the 6 criteria. Tells the wheel problem and the fix: the wheels slid and created friction, and less friction helped the robot move straight.

Missing: Why cost was left out, the winning score (23 of 30) and **why we chose HummerOne**. Check the wheel part: our log says one Technic ball joint, not two mini ball casters.

Next: Accomplished: finish the two blank questions and name the wheel part correctly. **Exceeds:** tell all three versions (original, mine cart, ball joint) with a reason for each and a before and after test result.

3 Our attachments (part 1)

CREATE

WHAT JUDGES WANT
For each attachment: what it does, what we changed from the original, and why. Do not only say what it is. Say what is new.

✎ **We built ___ attachments. They all attach to the robot's _____ using _____.**

✎ **Our attachments swap on and off in about ___ seconds. We tried _____ pins to make it faster, but ... so we went back to _____ because ...** (log entry 7)

✎ **We rated our attachments from 1 to 5 and chose _____ because ...**

Attachment: _____ **Mission(s):** _____

✎ **It started as a design from:**

✎ **What we changed from the original, and why:**

✎ **A problem we found, and how we fixed it:**

✎ **It is good at ... and not so good at ...**

Our attachments (part 2)

One box per attachment. Choose from: lifting arm with seed bucket, 2-axis arm with hook, forklift, Mission 1 attachment, Mission 11 basket.

Attachment: Forklift **Mission(s):** 9

It started as a design from:

Jack started building the forklift from Coach Zain. Ella and Amelia continued building it.

What we changed from the original, and why:

A problem we found, and how we fixed it:

We made a couple of mistakes, like when Sylvie moved up the gear bar. A problem Ella found was that certain things were built on the wrong side. Sylvie and others fixed this by going back in the instructions and rebuilding certain parts.

It is good at ... and not so good at ...

Attachment: Lifting Arm **Mission(s):** 2

It started as a design from:

The design it originated from was from a bot-fluencer.

What we changed from the original, and why:

We added a seed bucket because it couldn't bring the seeds back without one. We also added axle pins because it didn't stay on properly.

A problem we found, and how we fixed it:

The robot didn't line up with the mission. To fix this, we added what we call the masher. The masher forces/mashes the robot into place.

It is good at ... and not so good at ...

It is good when it does its job but it's not very consistent.

Review against the rubric (2026-11-12)

DEVELOPING **Forklift.** Good: an honest mistakes-and-fix story that names who worked on it. Missing: what it does for Mission 9, what we changed from the original, what it is good and not so good at, and test evidence ("we stabilized it"). Check the source: the draft says Coach Zain, the design log says Forklift1-zoptyk.

ACCOMPLISHED **Lifting arm.** Good: each change has a reason (seed bucket to bring seeds back, the masher to line the robot up), a problem and its fix, and an honest weakness. Missing: what it does in its mission (written as Mission 2, check), the bot-fluencer by name, and a number for "not very consistent" (how many out of 10?). Check: the design log says axle pins kept coming off and we went back to quick pins.

Next: Exceeds: give each attachment a test number and say what is new. Still to write: 2-axis arm and Mission 11 basket.

Our attachments (part 3)

Keep going until every attachment we will talk about has a box.

Attachment: The drone survey attachment **Mission(s):** 1

✎ **It started as a design from:**

The design was inspired by Team Turtlebots. We made one similar to theirs.

✎ **What we changed from the original, and why:**

We added some smaller curved pieces so it wouldn't get caught on things or on the mission.

✎ **A problem we found, and how we fixed it:**

The attachment used to be wiggly and loose. Our solution was to add extra points for it to attach to itself as well as some other pieces.

✎ **It is good at ... and not so good at ...**

It is good at flipping open the lidar map but it can't get the drone there.

ACCOMPLISHED **Mission 1 attachment (Drone Survey): review against the rubric (2026-11-12)**

Good: It says where the design came from (Team TurtleBots), what changed and why (smaller curved pieces so it does not catch), a problem and its fix (it was wiggly and loose, so we added attach points), and an honest limit (it flips the lidar map open but cannot get the drone there).

Missing: A test number (how many runs out of 10 does it open the lidar map?), our plan for the drone, and who built it. The design log lists three Mission 1 options; this is the TurtleBots one, so say we picked it and why.

Next: Exceeds: add the number, what is new compared with the TurtleBots design, and the plan for the drone.

Attachment: _____ **Mission(s):** _____

✎ **It started as a design from:**

✎ **What we changed from the original, and why:**

✎ **A problem we found, and how we fixed it:**

✎ **It is good at ... and not so good at ...**

4 Our code and sensors (part 1)

CREATE

WHAT JUDGES WANT
Explain it step by step, as if the judge has never coded. Say what each sensor does for us.

✎ **We use ___ sensors on our robot: _____. The sensor in the _____ does _____, and we use it to _____.**

We use 2 colour sensors and 1 gyro sensor. The colour sensors sense the colours on the mat. We use them for squaring. The gyro sensor ...

✎ **We added a second color sensor because ...**

The reason behind why we added a 2nd colour sensor is because squaring is when one sensor sees the colour black, it will turn until the other sensor sees black. It will then wiggle till straight. It drives crooked without squaring.

✎ **Squaring up on a line, step by step:** (with one sensor it did not work. With two, what happens when one sees black?)

First, ...

It drives up to the line diagonally and senses the black line.

Then, ...

It turns until the other sensor senses the black line.

Finally, ...

It wiggles until it is straight.

ACCOMPLISHED **Review against the rubric (2026-11-12)**

Good: The squaring steps are clear and in order, and the reason for a second sensor is clear. Sensors are named and each has a job.
Missing: The gyro sentence is unfinished. Where is each sensor mounted? There is no proof yet that squaring works.
Next: Exceeds: finish the gyro sentence, then add before and after proof (for example, how crooked it ended with and without squaring).

4 Our code and sensors (part 2)

CREATE

WHAT JUDGES WANT

Gyro Move, Kp and Ki, gyro turns, the speed ramp and the centimetre conversion. See log entries 8 and 10.

✎ **To drive straight our code uses a block called _____. Every 0.1 seconds the robot checks _____ and if it is off it ...**

Our code uses a block called Gyro Move. Every 0.1 seconds the robot checks the yaw angle and if it is off it corrects itself.

✎ **Kp decides _____. Ki decides _____.**

Kp decides how it corrects itself over a short period of time. Ki decides how it corrects itself over a long period of time.

✎ **For turning we use _____ instead of motor degrees because ...** (yaw 90 versus motor 90)

Yaw angle. Yaw is more reliable. We also increase the speed of the turn depending on how big the turn is.

✎ **The robot starts slowly, speeds up and slows at the end because ...**

That makes it so that the robot doesn't go farther than we want it to.

✎ **We changed the distance block from _____ to _____ because ... We measured that ___ motor degrees is ___ cm by ...**

From degrees to cm, because (degrees are) not as easy to measure than (cm).

DEVELOPING **Review against the rubric (2026-11-12)**

Good: A simple but correct explanation: Gyro Move checks the yaw angle and corrects itself, Kp and Ki are short and long term correction, turns use yaw because it is more reliable, the ramp stops the robot going farther than we want, and distances are in cm because that is easier to measure.

Missing: Step by step detail and numbers. Why is yaw more reliable (a 90 degree yaw turn is a real 90 degree robot turn, a 90 degree motor turn is only a quarter of a wheel turn)? What are our values (Kp 2, Ki 1)? How did we measure cm (2070 motor degrees = 100 cm, which is 20.7 degrees per cm)? Check the turn speed: our log says the turn slows as it nears the angle.

Next: Accomplished: add the why for yaw, the numbers for Kp, Ki and cm, and one example of a Gyro Move step by step.

Exceeds: add how this made the robot repeat (the 1 m drive).

5 Testing and improving (part 1)

ITERATE

WHAT JUDGES WANT

This is where teams exceed: **repeated** testing and improvements **because of** the testing. Use numbers. Pick the story you have the best evidence for.

Ideas: ☐ Kp and Ki tuning ☐ 1 m drive, 10 times ☐ seed basket (3, 5, 7 out of 10) ☐ front wheel changes ☐ quick versus axle pins ☐ forklift tipping ☐ squaring with two sensors

✎ **Story 1. We wanted to know ...**

If we had a good Kp and Ki.

✎ **So we tested by _____, __ times. The results were: (give numbers)**

Kp and Ki, 15 times. The result was that the Kp was already good. When we tested it at 1 the avg error was 26 mm, we tested 3 and the error was 13. Our original Kp of 2 was under 1 mm error.

✎ **That told us ... So we changed ...**

We tested Ki at 0.5, 1, 2. We found changing it from 2 to 1 we were more accurate by 5 mm.

✎ **After the change we tested again and ...**

It worked very well.

ACCOMPLISHED **Review against the rubric (2026-11-12)**

Good: Clear repeated testing with numbers: Kp values tested several times each, Kp 1 about 26 mm off, Kp 3 about 13 mm off, Kp 2 under 1 mm. Then Ki 0.5, 1 and 2, and the change from Ki 2 to 1 made us about 5 mm more accurate. The change comes from the test results.

Missing: Plain words for what Kp and Ki do. "It worked very well" needs a number: our old Kp 2.1 and Ki 2 was about 9.6 mm off, and Kp 2 with Ki 1 ended within 1.6 mm. The "15 times" looks like 5 Kp values times 3 runs (check).

Next: Exceeds: add the before and after number, say the sample was small so we planned a 10 run check (the 1 m drive), and say what Kp and Ki are for a judge who has never coded.

5 Testing and improving (part 2)

ITERATE

WHAT JUDGES WANT
A second test story with numbers. Then be honest about what is not finished and say your plan.

Story 2. We wanted to know ...

If we could complete our missions in the time limit.

So we tested by _____, __ times. The results were: (give numbers)

We tested it by doing the missions with a stopwatch repeatedly.

That told us ... So we changed ...

So we added blocks with a timer so the robot would carry on instead of stopping and wasting tokens.

After the change we tested again and ...

We wasted less precision tokens and got more points in the end.

Not finished yet: we still need to _____, and our plan is to _____. (saying this sounds confident, not weak)

We still need to work on some funky code.

DEVELOPING **Review against the rubric (2026-11-12)**

Good: A clear test story: we wanted to know if the missions fit the time limit, tested with a stopwatch, added timer blocks so the robot keeps going instead of stopping and wasting tokens, and scored more points. It says what is not finished.
Missing: Numbers. How many runs? What times and points before and after? How many precision tokens? Say what a precision token is. "Some funky code" needs a name and a plan.
Next: Accomplished: use the numbers from the timing sheet (8 runs, best 185 points in 2:50) and say which runs were before and after the timer blocks. **Exceeds:** name the code to fix and how it gets us under 2:30.

6 What we learned

COMMUNICATE

WHAT JUDGES WANT
A detailed explanation of the process and the lessons we learned. Mistakes that taught us something are good here.

The biggest thing we learned about building a robot is ...

The biggest thing we learned about coding is ...

Something we thought would work but did not was _____. We learned ...

If we were giving advice to a new team we would tell them ...

If we had more time we would ...

NOT WRITTEN YET **Review against the rubric (2026-11-12)**
Good: Nothing written yet.
Missing: Lessons, a mistake we learned from, advice for a new team.
Next: The forklift mistakes (wrong side, gear bar) and the wheel problem are good lessons.

7 Teamwork and closing

DESIGN + COMMUNICATE

WHAT JUDGES WANT

All team members contributed ideas, with building **and** coding skills in all members. Plus pride and enthusiasm, and a Core Value.

Name	One thing I built or designed	One thing I coded, tested or learned
Sylvie		
Brynn		
Jack		
Fedor		
Ira		
Jordan		
Ella		
Alexis		
Amelia		

✎ **Core Value moment (teamwork, discovery, inclusion, innovation, impact or fun): when _____ ...**

✎ **The part of our robot we are proudest of is _____ because ...**

✎ **Thank you for listening. We would love to answer your questions about ...**

NOT WRITTEN YET **Review against the rubric (2026-11-12)**

Good: Nothing written yet. Teamwork already shows in the forklift story: Jack, Ella, Amelia and Sylvie.
Missing: One build thing and one coding or testing thing for all 9 of us.
Next: Fill the table, pick a Core Value moment, and write the closing.

Before we practice

	Checklist
☐	Every blank has a fact, a number or an [ASK: name]
☐	Every choice has a "because"
☐	All 9 teammates speak
☐	At least 2 test stories with numbers
☐	For each attachment: what we changed from the original
☐	Code explained step by step
☐	A lesson from a mistake, and a Core Value
☐	Read out loud, under 5 minutes, without the field and without reading from the page

Questions judges may ask

Jot a short answer or the facts you will use.

1. Why did you choose this base robot?

2. What is the most important change you made, and why?

3. How do you know your robot is accurate?

4. Walk us through how your code works.

5. What did you do when something did not work?

6. Who built each attachment? Who wrote the code?

7. How did you choose which attachment for each mission?

8. What would you change with more time?